

Rishi Bala

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Education

The University of Texas at Austin, Austin, Texas

August 2025 – May 2029

- GPA: 3.97/4, BS: Electrical and Computer Engineering Honors, BBA: Canfield Business Honors, ACT: 36/36
- Relevant Coursework:** Computer Architecture, Digital Logic Design, Circuit Theory, Introduction to Embedded Systems

Experience

IPC Systems | New York, New York

May 2026 – August 2026

Embedded Firmware Intern

- Replaced polling-based turret screenshot delivery with live streaming by benchmarking VNC and WebRTC pipelines on embedded Linux BSP hardware, measuring end-to-end frame latency to drive a production protocol decision for financial trading floors.
- Instrumented the embedded turret diagnostics pipeline to surface real-time performance telemetry across trading floor endpoints.
- Extended the Rust/Tauri hardware interface with WAMP command surfaces, bridging the turret BSP to a desktop viewer.
- Collaborated with senior engineers to architect a hardware-software stack spanning BSP daemons (C/C++, GStreamer, Wayland), encrypted transport, and the Rust viewer application, enabling integration across embedded and desktop environments.

Longhorn Racing – Electric | Austin, Texas

August 2025 – Present

Controls System Lead (Competing in the Formula SAE International Competition)

- Implemented real-time torque arbitration, torque vectoring, and traction control with deterministic CAN I/O and BSPD safety logic.
- Built STM32 embedded control and telemetry systems with CAN networking, multi-sensor interfacing, and real-time firmware.
- Integrated safety interlocks, fault detection, and watchdog logic across all vehicle subsystems for competition reliability.
- Designed an automotive-grade wiring harness, including connector selection, crimping, CAN routing, and full-vehicle integration.
- Coordinated cross-functional integration with mechanical and high-voltage subteams, validating 250+ miles of vehicle testing before competing at the Formula SAE International Competition.

Projects | www.rishibala.com

FPGA-Based CME Market Making Engine

April 2026 – June 2026

[GitHub Repository](#)

- Designed a fully pipelined, CPU-free market maker in SystemVerilog on a Xilinx Artix-7 with a complete Ethernet/TCP/IP network stack, CME MDP 3.0 market data, and iLink 3 FIXP order entry across 25 RTL modules; sub- μ s logic-pipeline tick-to-order latency.
- Implemented a CME MDP 3.0 market data pipeline decoding SBE-encoded UDP multicast packets across incremental and snapshot feeds, with a 10-level order book in RTL featuring gap detection and automatic snapshot-based sequence recovery.
- Built a quoting engine computing VWAP mid-price via a pipelined 64-bit integer divider, adjusting bid/ask quotes for inventory skew and an exponentially-decaying Order Flow Imbalance (OFI) signal derived from real-time aggressor trade data.
- Built FSMs for all state-bearing modules, FIFOs for inter-stage flow control, and toggle-based CDC synchronizers for domain crossing.

Vehicle Control Unit (VCU)

July 2026 – Present

- Architected VCU safety logic for FSAE EV rules compliance with BSPD, shutdown circuit sequencing, and APPS plausibility checks.
- Arbitrated torque requests from APPS, regen, and torque vectoring inputs with priority logic for driver intent and thermal limits.
- Maximized circuit protection with TVS clamping, series resistance, current limiting, and fusing across all analog VCU interfaces.
- Designed PCB schematics and layout in KiCad, applying signal integrity and EMC constraints for CAN bus routing.

Thermal Sensor Module (TSM)

August 2025 – April 2026

[GitHub Repository](#) | [Hardware Pictures](#)

- Led the development of the TSM, a module combining radiator fan control with integrated cooling loop and flow rate monitoring.
- Owned the entire design process, including component selection, schematic design, soldering, testing, and firmware development.
- Wrote latency-optimized firmware using FreeRTOS on top of an STM32 microcontroller; developed a CAN-based telemetry system.

YerraGlasses

December 2025 – March 2026

[GitHub Repository](#) | [Hardware Pictures](#)

- Designed ESP32-S3 audio-augmented smart glasses with dual I2S, real-time ML object detection, and low-latency auditory feedback.
- Developed a space-constrained PCB with USB-C power delivery and rail regulation for a low-latency vision-to-audio pipeline.

Additional Information and Skills

Technical Skills: SystemVerilog, Verilog, FPGA, Vivado, UVM, C/C++, Rust, RTOS, STM32, CAN, PCB Design, AXI4-Lite, CDC

Honors: University Honors 2x, VEX Robotics World Finalist, World Division Champion, National Merit Commended Scholar

Interests: Pickleball, Motorsport, Drums, Jazz, Tennis, Go-Karting

Work Eligibility: Eligible to work in the U.S. with no restrictions